

A vibrant blue sky filled with fluffy white clouds. A bright sun is positioned in the upper center, creating a strong lens flare effect. The text 'Cloud Native WebAssembly' is centered in the middle of the image.

# **Cloud Native WebAssembly**

# Cloud Native

=

build

deploy

manage



aws



Public clouds



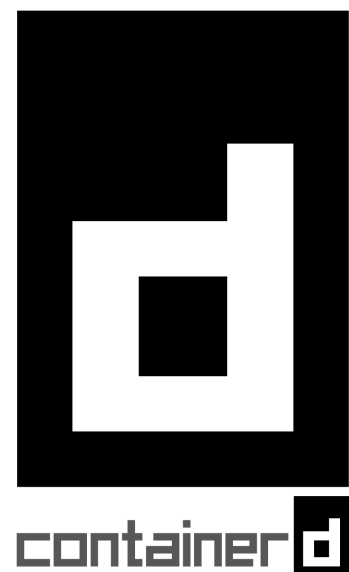
Private clouds



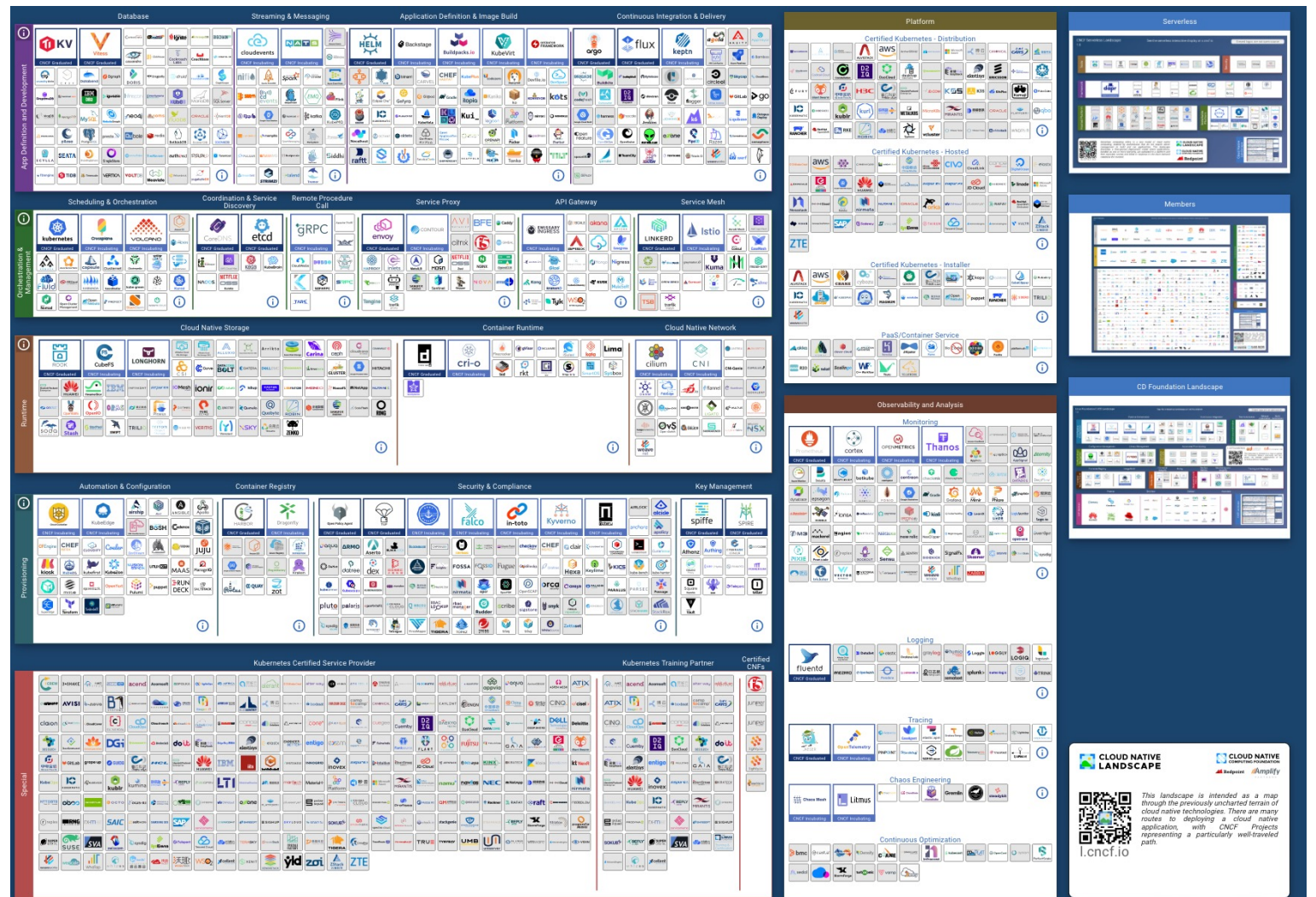
aws



Hybrid clouds

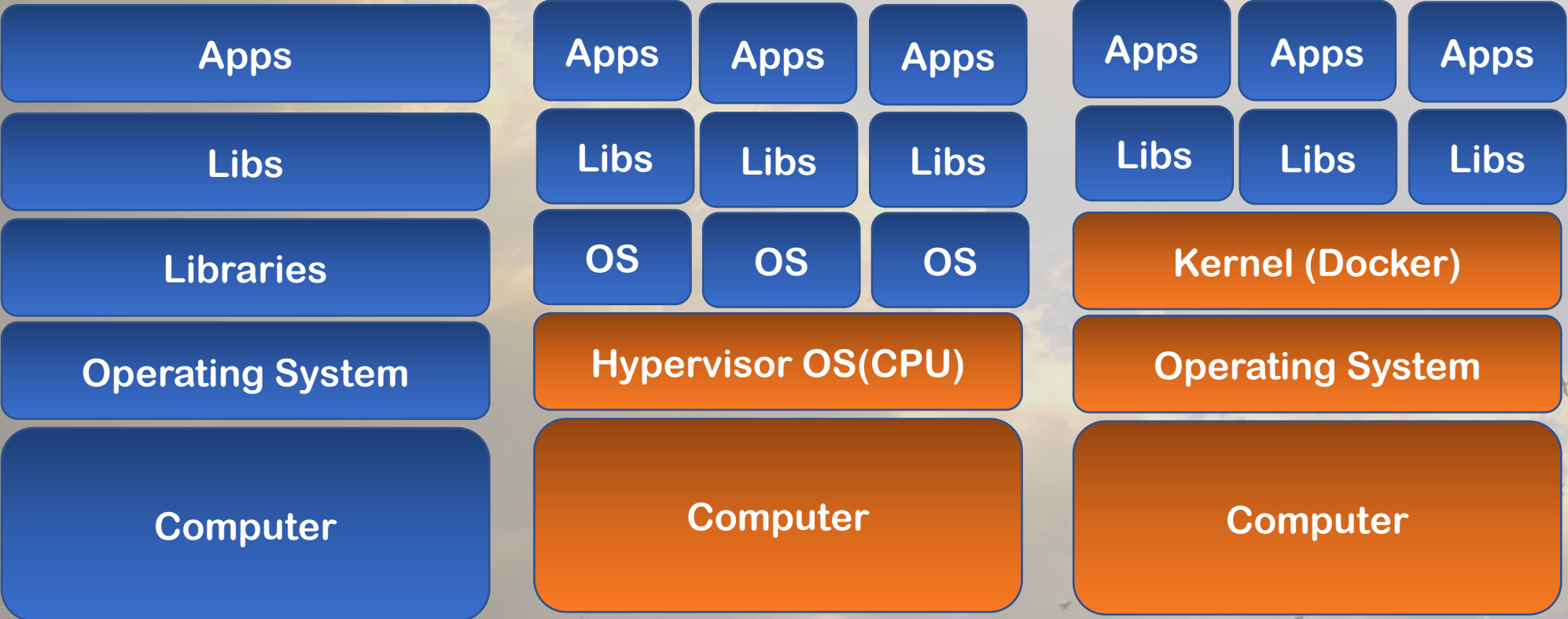
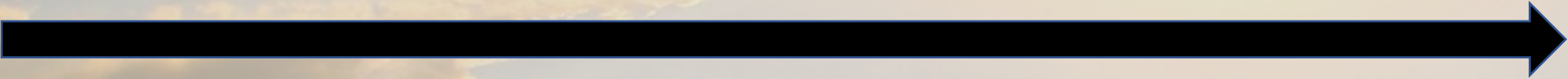


# The CNCF landscape



**Tightly  
coupled**

**Declarative**



**Datacenter**

**Public cloud**

**Containers**

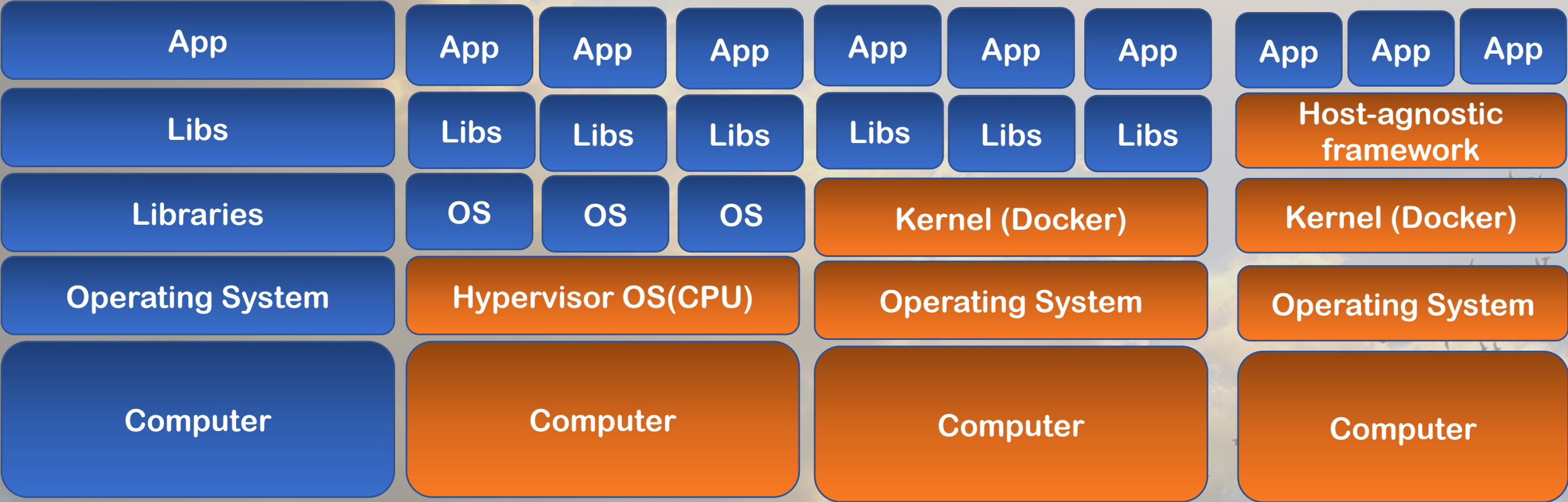
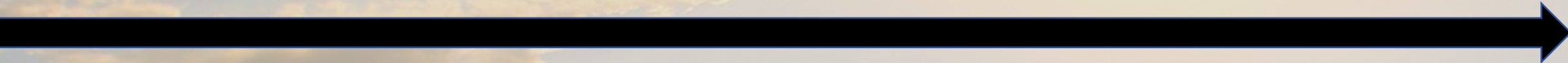




Where are we headed?

**Tightly  
coupled**

**Declarative**

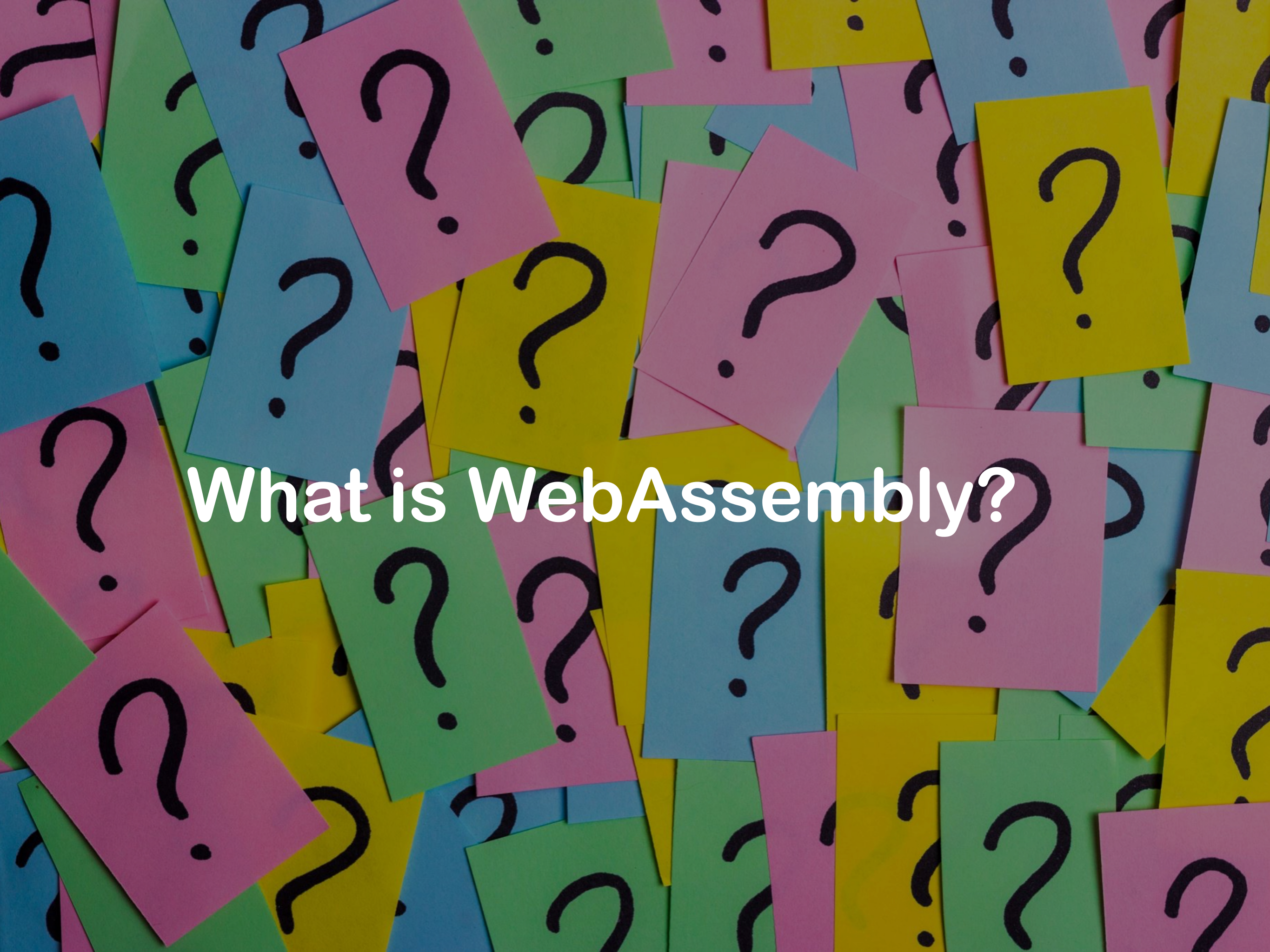


**Datacenter**

**Public cloud**

**Containers**

**Write once, Run  
anywhere**

The background of the image is a dense, overlapping collage of numerous small, rectangular sticky notes. These notes are in four distinct colors: pink, yellow, light blue, and light green. Each sticky note features a large, bold, black question mark. The notes are scattered across the entire frame, creating a textured and busy visual effect. In the center of this collage, the text "What is WebAssembly?" is written in a clean, white, sans-serif font.

What is WebAssembly?



“WebAssembly (abbreviated *Wasm*) is a binary instruction format for a stack-based virtual machine. Wasm is designed as a portable compilation target for programming languages, enabling deployment on the web for client and server applications.”



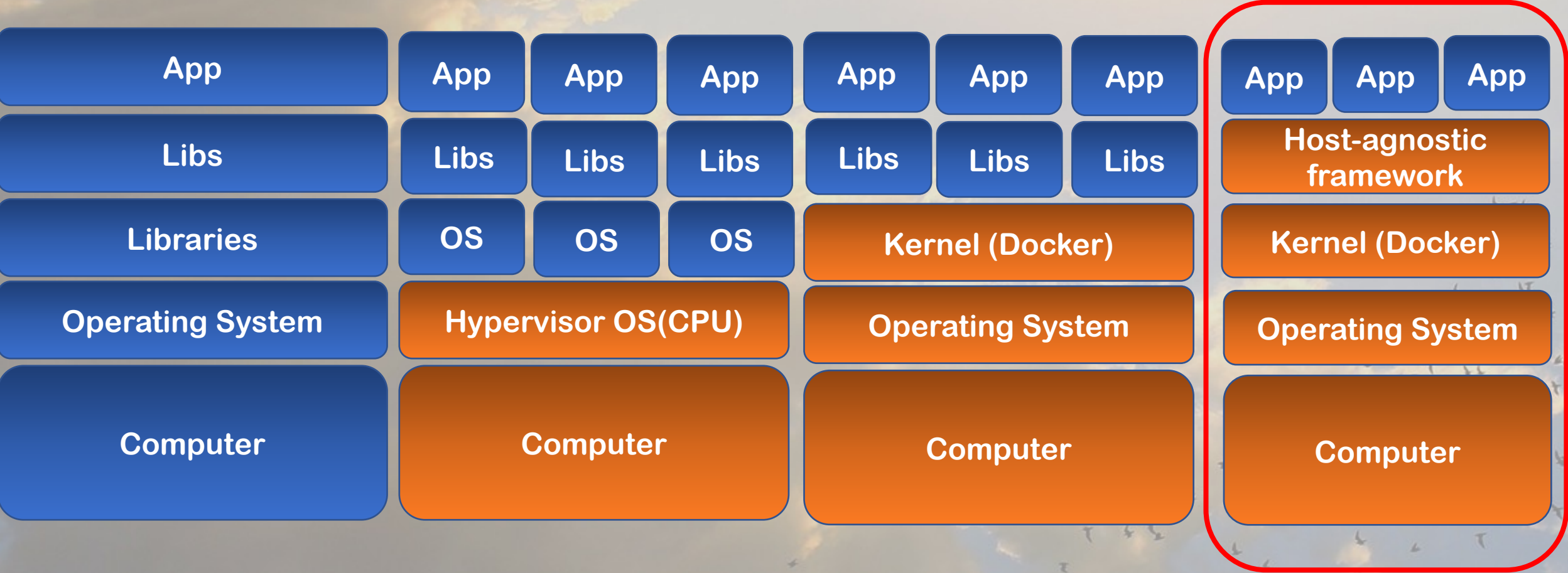
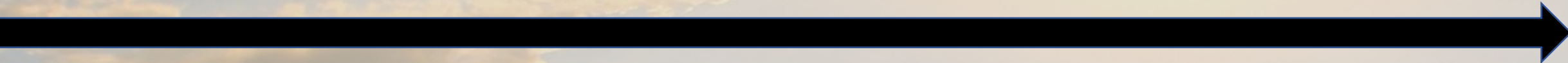
**Does this  
mean  
Wasm is  
ONLY for  
the web?**



# Cloud Native ❤️ Wasm?

**Tightly coupled**

**Declarative**



**Datacenter**

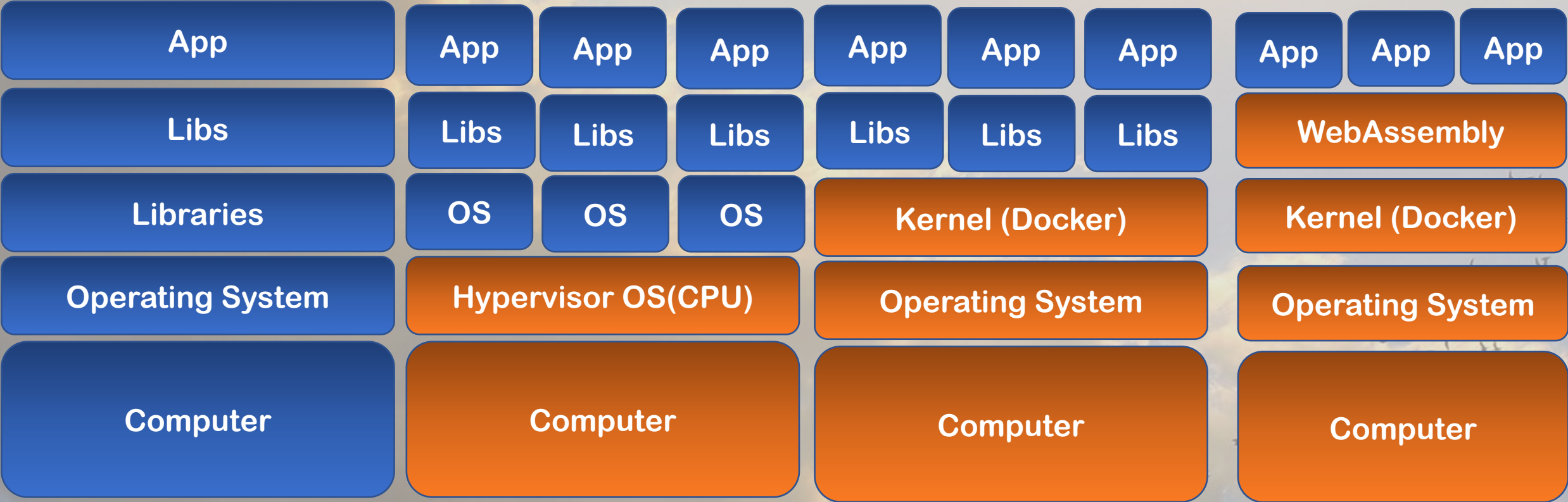
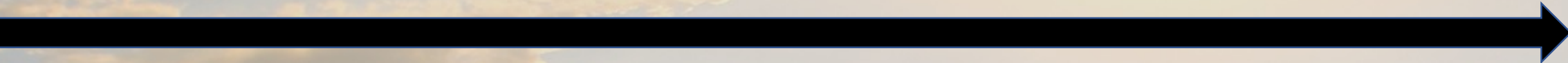
**Public cloud**

**Containers**

**Write once, Run  
anywhere**

**Tightly  
coupled**

**Declarative**



**Datacenter**

**Public cloud**

**Containers**

**Integrated WebAssembly**

# What Makes Wasm “Cloud-native” worthy?



PORTABILITY



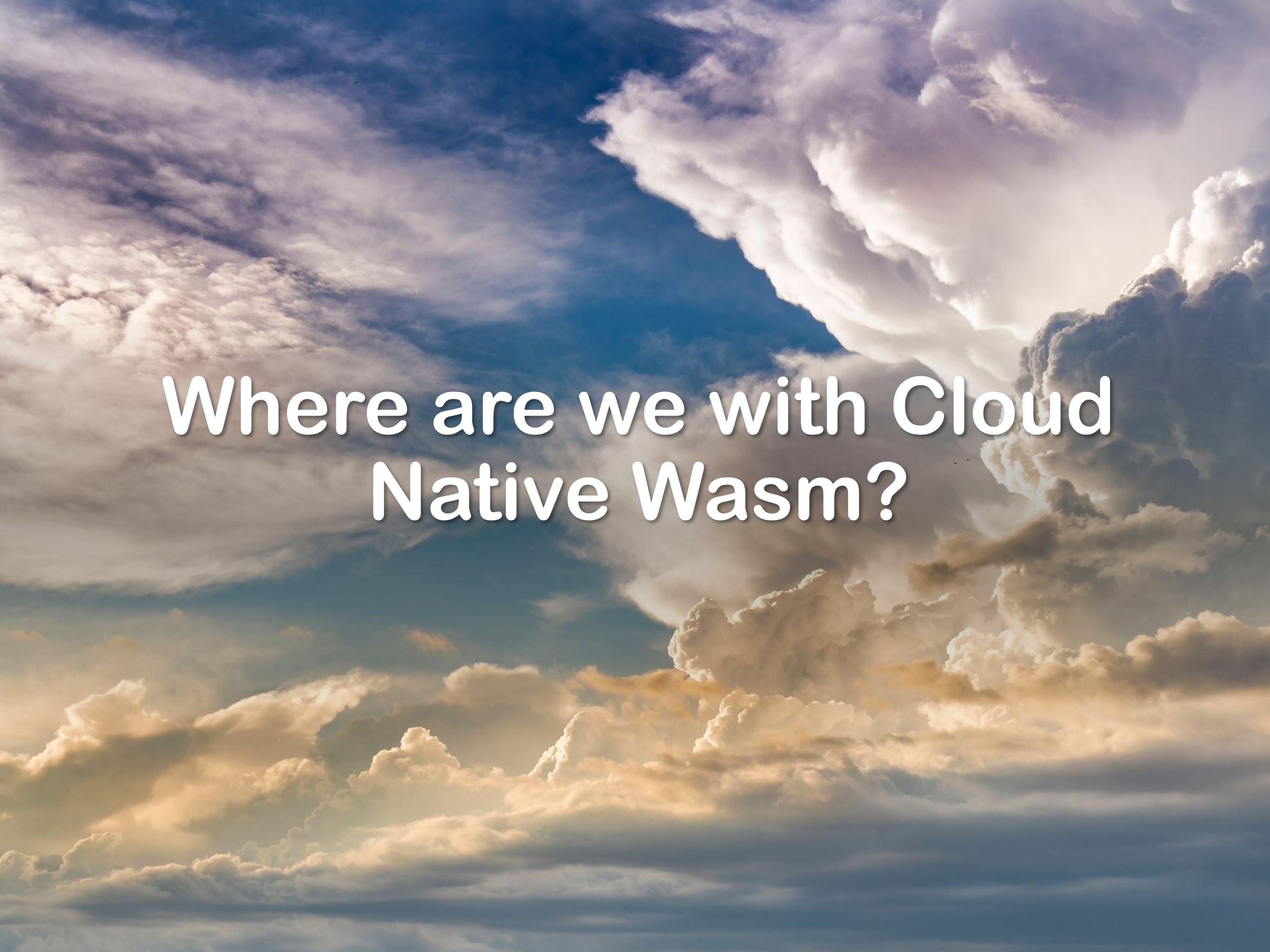
SPEED



SIZE

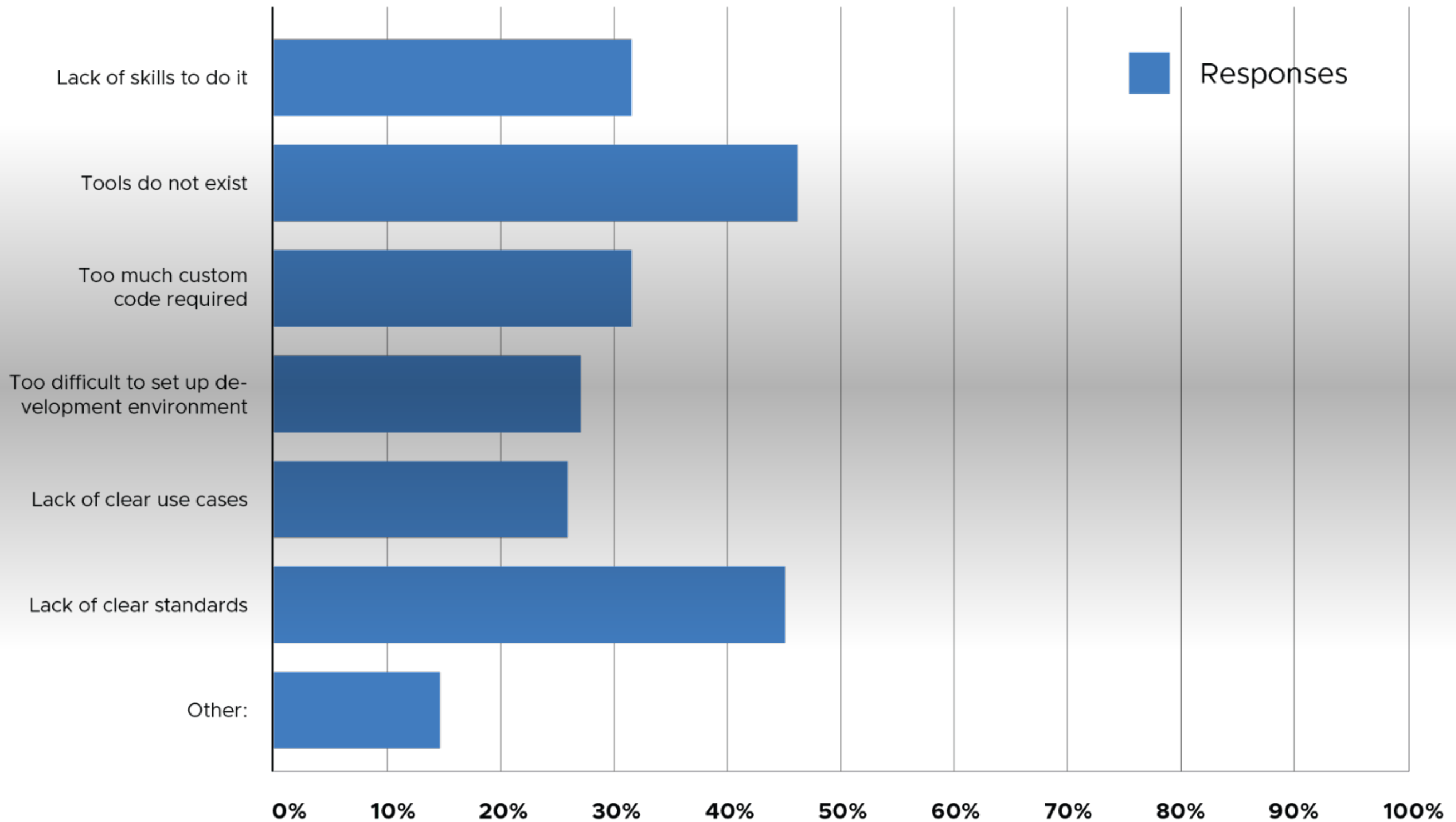


SECURITY

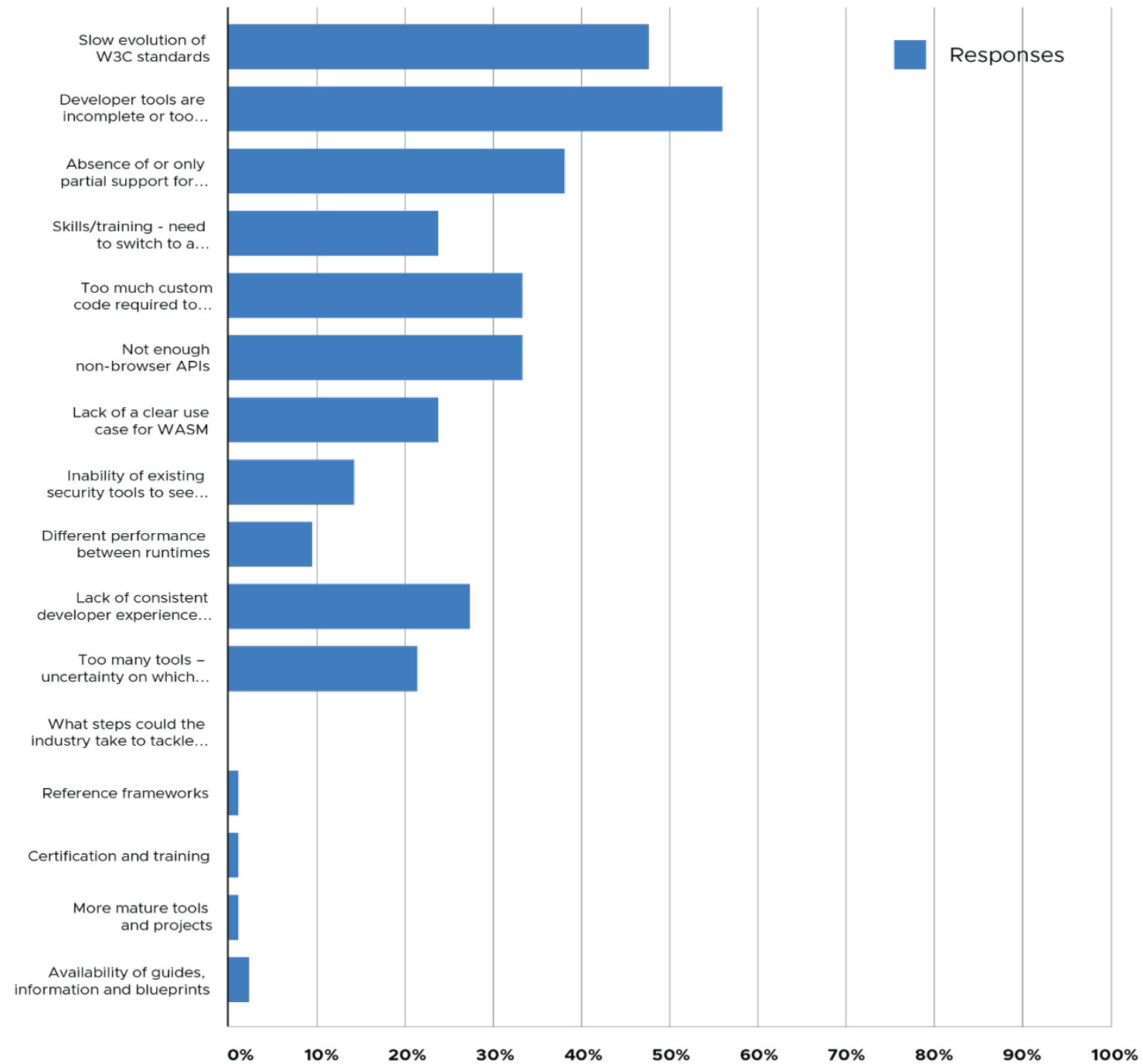


Where are we with Cloud  
Native Wasm?

# What are the biggest barriers you face when using WASM or in wanting to use it?



## What concerns do you have about WASM? Tick all that apply:



The image features three red location pins of varying sizes on a blue, textured surface. A winding road with a dashed white line runs through the scene. The text 'Where do we go from here?' is overlaid in white with a drop shadow. The background is a soft-focus blue sky with a bright light source in the upper right.

Where do we go from here?

# The way ahead

- Onboarding of official support for Wasm in compilers for all major languages
- Evolution of the ecosystem
  - The Component Model
  - Various proposals
    - Garbage Collection
    - Exception Handling
    - Memory with 64-bit indexes

# The way ahead

- Easier package management
  - OCI-compliant registries can store artifacts other than container images as of Sep 2022.
    - Includes Wasm applications
    - WIP to make this a reality!
- Transforming the serverless landscape

How do I  
get  
started?

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